

Code-Layout,ReadabilityandReusability

Date	15March2024
TeamID	XXXXXX
ProjectName	Smart Lender-Applicant Credibility Prediction for Loan Approval
MaximumMarks	5 Marks

Code-Layout,ReadabilityandReusability:

The SmartLender project follows a clean and organized code structure with proper file organization, meaningful variable and function names, and consistent indentation. The code is divided into reusable modules for data preprocessing, model training, prediction, and the Flask web application. Comments and basic error handling improve readability, making the project easy to understand, maintain, and extend.

CodeLayoutChecklist:

S.No	CodeQualityParameter	Description	Followed (Yes/No/Partial)	Remarks
1	ConsistentIndentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained in Python and Flask files.
2	ProperFileStructure	Files and folders are logically organized	Yes	Separate folders for model, templates, static files, and application code.
3	MeaningfulVariableName	Variables reflect their purpose clearly	Yes	Variables such as <code>loan_amount</code> , <code>credit_history</code> , and <code>income</code> are descriptive.
4	Function/MethodNames	Functions are descriptively named	Yes	Functions like <code>predict_loan()</code> and <code>load_model()</code> clearly indicate their purpose.
5	CodeComments	Inline and block comments explain logic	Partial	Comments are added for important sections; more comments can improve readability.
6	ModularDesign	Code is split into reusable functions/modules	Yes	Machine learning model, Flask app, and preprocessing are organized into modules.

7	NoRedundantCode	Duplicateorunused codeisremoved	Yes	Repeated code is minimizedandreusable functions are used.
8	ErrorHandling	Exceptionsanderrorsarehandled gracefully	Partial	Basicinputvalidation is implemented; additional exception handling can be added.

ReusableComponents/Modules:

S.No	Component/Module Name	Language/ Technolo	yWhereReused	Reusability Level (High/Medium/Low)
1	Data Preprocessing Module	Python, Pandas	Data cleaning and feature preparation	High
2	ML Prediction Model (XGBoost)	Python, Scikit-learn, XGBoost	Loan approval prediction	High
3	Flask Web Application	Python, Flask	User input handling and prediction	High
4	HTML/CSS User Interface	HTML, CSS	Input formand result page	Medium
5	Model Loading Module	Python (Joblib/Pickle	Loading trained model for prediction	High

OverallCodeQualityAssessment:

Aspect	Rating(1-5)	Comments
CodeLayout &Structure	5	Well-organized project with separate folders for models, templates, and static files.
Readability	5	Uses meaningful variable names and proper indentation.
Reusability	5	Modular functions for preprocessing, training, and prediction can be reused.
Documentation/ Comments	4	Basic comments are provided;more detailed comments can be added.
OverallScore	4.8 / 5	Clean, maintainable, and reusable code suitable for deployment.